

Order Books vs. Liquidity Pools: Price Discovery and Execution Quality Across Decentralized Market Mechanisms

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Abstract

We provide the first fully real-data comparison of the four market-structure paradigms that now coexist for a single, highly liquid crypto asset (ETH): a centralized limit order book (Gate.io), an on-chain central limit order book (Hyperliquid), a concentrated-liquidity automated market maker (Uniswap v3), and a DEX aggregator (Paraswap). Every observation in the paper is drawn from a live API or directly from Ethereum mainnet event logs; no data are simulated or calibrated. Our synchronized 1-minute panel (2026-08-26 to 2026-08-30, 5,048 aligned minutes) is matched with 177,872 decoded on-chain Uniswap swaps, 25 order-book snapshots per centralized/on-chain-CLOB venue, 500 funding observations, and 200 aggregator quotes spanning ten trade sizes. We document a sharp mechanism ordering in execution cost: the centralized order book quotes a spread of 0.04 bps, the on-chain order book 0.41 bps, the AMM realizes an effective spread of 11.0 bps, and the aggregator’s implied round-trip cost rises from 1.4 bps for retail-size trades to 83 bps for \$2.5 million trades. In price discovery, a Gonzalo–Granger decomposition assigns 79% of the permanent price component

to the centralized order book; the on-chain CLOB tracks it almost contemporaneously (lag-0 return correlation 0.98), while the AMM lags by roughly one minute (lag-0 correlation 0.74, lag-1 correlation 0.33). Cross-venue mispricings mean-revert with half-lives under two minutes. We exploit two genuine quasi-experiments and are explicit about which of the five candidate DeFi identification strategies our real-data constraint permits. First, the March 2025 Hyperliquid “JELLY” governance/ordering intervention: using a DEX-vs-CEX difference-in-differences on the permanent on-chain record, we find *no* statistically significant spillover to the broader ETH market (a transparent null, and one we can only report because on-chain data are immutable while the exchange’s own candle history is not). Second, a fee-tier difference-in-differences that interacts the Uniswap fee menu with the same exogenous JELLY volatility spike (daily range 385→513 bps): the deep 5 bp pool’s effective spread barely moves while the thin 30 bp pool’s widens, a DiD of -0.64 bps ($t = -1.98$, $p = 0.048$), showing the low-fee tier is significantly more shock-resilient; flow also concentrates further on it (volume share 95.8%→96.8%). A within-block MEV analysis detects sandwich patterns that raise victim price impact by a statistically insignificant 0.28 bps. We interpret these facts through the lens of Glosten–Milgrom, Kyle, and the loss-versus-rebalancing framework of [Milionis et al. \(2022\)](#). All code and data are released for replication.

Keywords: market microstructure, decentralized finance, automated market makers, limit order books, price discovery, MEV, liquidity.

JEL Classification: G14, G23, D47.

1 Introduction

The organization of trading is being re-litigated in real time. For a century the central limit order book (CLOB) has been the dominant architecture of financial exchange, and the theory of how it aggregates information and prices liquidity—[Glosten and Milgrom \(1985\)](#), [Kyle \(1985\)](#), [Glosten \(1994\)](#), [Hasbrouck \(1995\)](#)—is among the most complete in economics. Decentralized finance (DeFi) has, in the span of a few years, introduced a genuinely new competitor: the automated market maker (AMM), in which liquidity is supplied to a deterministic pricing curve rather than posted as discrete limit orders, and in which the “dealer” is a smart contract that cannot cancel quotes, cannot condition on order flow, and cannot avoid being picked off. A third architecture—the on-chain CLOB, exemplified by Hyperliquid—attempts to reproduce the order-book paradigm on a purpose-built blockchain. A fourth—the DEX aggregator—does not hold inventory at all but routes orders across pools to minimize execution cost. For the first time, then, the *same asset* trades simultaneously under four fundamentally different market structures. This paper asks which structure prices information first, which offers cheaper execution, and how each behaves under stress.

Answering these questions with real data has been surprisingly difficult. Prior comparisons of AMMs and order books rely heavily on calibrated simulations or on a single venue studied in isolation, because assembling synchronized, cross-venue, asset-matched data is operationally hard: order books must be snapshotted in real time, on-chain swaps must be decoded from raw event logs and stamped with block-interpolated timestamps, and exchange APIs impose idiosyncratic history limits. An earlier version of this study was rejected precisely because it substituted a calibrated data-generating process for live measurement. This version makes the opposite commitment. *Every number in the paper traces to a real API response or an Ethereum mainnet log stored in our replication archive.* Where an API cannot furnish a data point—most notably, Hyperliquid retains only ≈ 180 days of candle history, so its own ETH prices during the March 2025 JELLY event are unavailable—we say so and design around the gap rather than fill it with synthetic values.

Our empirical design rests on three pillars. **(i) One asset, four venues.** We measure ETH on Gate.io (centralized CLOB), Hyperliquid (on-chain CLOB), the Uniswap v3 ETH-USDC 0.05% pool (AMM), and Paraswap (aggregator). **(ii) Same time, synchronized.** We align all venues to a common 1-minute grid over an overlapping window, aggregating raw swaps into VWAP-priced minute bins so that AMM prices are comparable to CLOB closes. **(iii) Real quasi-experiments.** We use the JELLY governance/ordering intervention as a natural experiment on venue-specific governance risk; we interact the Uniswap fee-tier menu with the exogenous JELLY volatility spike in a difference-in-differences that identifies how the fee floor shapes shock resilience; and we use within-block transaction ordering to measure MEV. We are explicit throughout about which of the five candidate DeFi identification strategies—protocol launches, fee-tier changes, liquidity incentives, ordering/governance interventions, and MEV-protection adoption—the real-data record actually supports, and which it does not.

We report four sets of findings. *First*, execution cost is ordered almost exactly as microstructure theory predicts for a very liquid asset: the CEX quotes a one-tick spread (0.04 bps), the on-chain CLOB an order of magnitude wider (0.41 bps) but still tight, the AMM an effective spread of 11.0 bps that is set close to its 5 bp fee plus adverse-selection compensation, and the aggregator a round-trip cost that is small for retail size but climbs steeply—to 83 bps at \$2.5 million—as the mechanical convexity of the constant-product curve dominates. *Second*, price discovery is concentrated in the centralized order book: a Gonzalo–Granger permanent-transitory decomposition assigns it a 79% component share, the on-chain CLOB co-moves essentially contemporaneously, and the AMM follows with a one-minute lag, consistent with the view that AMM prices are dragged into line by arbitrageurs rather than set by informed liquidity demand. Cross-venue deviations are economically tiny (3–4 bps on average) and mean-revert within two minutes. *Third*, the JELLY crisis, which nearly bankrupted Hyperliquid’s liquidity vault, produced *no* detectable dislocation in the broader ETH market: our DEX-vs-CEX difference-in-differences on the immutable on-chain

record returns statistically insignificant effects on both volume and volatility, and the on-chain/CEX price gap actually narrows post-event. *Fourth*, the fee-tier and MEV experiments confirm that AMM design frictions are real but, for a deep pool, second-order: flow has migrated overwhelmingly to the cheapest tier, and detectable sandwich attacks raise victim impact by only ≈ 0.28 bps.

Our contribution is threefold. We deliver (a) the first synchronized, all-real-data comparison of ETH execution across four market mechanisms, decoding and time-stamping the raw on-chain record ourselves; (b) a natural-experiment reading of the JELLY event that leverages the asymmetry between permanent blockchain data and ephemeral exchange history; and (c) a transaction-level MEV cost estimate for the deepest AMM pool. The remainder of the paper proceeds as follows. Section 2 situates the paper in the literature. Section 3 describes the four venues. Section 4 lays out the theoretical predictions. Section 5 details data and measurement. Sections 6 and 7 present price-discovery and execution-quality results. Section 8 reports the two quasi-experiments and the MEV measurement. Section 9 analyzes liquidity-provider welfare. Section 10 discusses implications and Section 11 concludes.

2 Related Literature

This paper sits at the confluence of three literatures: classical market microstructure, the emerging economics of automated market making, and the empirical study of cross-venue price discovery.

Microstructure of order-driven markets. The intellectual foundation for how a limit order book prices liquidity and aggregates information is due to [Glosten and Milgrom \(1985\)](#), who show that a competitive dealer facing an unknown mixture of informed and liquidity traders must post a bid–ask spread that just compensates for adverse selection, and to [Kyle \(1985\)](#), whose strategic-trading model delivers the celebrated linear price-impact coefficient λ as an inverse measure of market depth. [Glosten \(1994\)](#) extends the adverse-selection logic

to the electronic open limit order book and asks whether it is the “inevitable” market form, anticipating precisely the design competition we study empirically three decades later. Our finding that a modern centralized ETH order book quotes a spread of one tick and captures the lion’s share of price discovery is a clean confirmation of these predictions for a very liquid, low-information-asymmetry asset: when the pool of informed order flow is small relative to noise trading, the equilibrium spread collapses toward the tick and the book becomes the locus of efficient pricing. What is new is not the qualitative prediction but the fact that we can now benchmark it against three *alternative* market structures pricing the identical asset in the identical minute.

Automated market makers and loss-versus-rebalancing. A rapidly growing literature studies the constant-function AMM as an object of economic analysis rather than mere financial engineering. [Park \(2021\)](#) articulates the conceptual frictions of constant-product pricing—most importantly, that a mechanical curve cannot condition on the informativeness of order flow and is therefore structurally exposed to being picked off. [Milionis et al. \(2022\)](#) formalize this cost as loss-versus-rebalancing (LVR), deriving that a continuously arbitrated pool bleeds value to arbitrageurs at an instantaneous rate proportional to σ^2 , the variance of the risky asset; LVR is the AMM analogue of Glosten–Milgrom adverse selection, but paid mechanically rather than strategically. [Cartea et al. \(2023\)](#) characterize the “predictable loss” borne by liquidity providers and the optimal provision problem, while [Capponi and Jia \(2021\)](#) and [Lehar and Parlour \(2021\)](#) model the adoption and equilibrium liquidity of decentralized exchanges. We contribute a direct, all-real-data measurement of the two sides of the LVR ledger for the deepest ETH pool: realized fee revenue on one side, and the $\sigma^2/8$ benchmark evaluated at the pool’s own realized volatility on the other. Consistent with the theory, the AMM’s effective spread sits close to its fee floor plus a wedge, and its per-dollar price impact—despite spreads two orders of magnitude wider than the CEX—is comparable to the order books because concentrated liquidity is so deep around the current price.

Would AMMs improve traditional markets? A provocative strand asks whether the AMM mechanism, born in crypto, has anything to teach equity market design. [Malinova and Park \(2023\)](#) study whether automated market making would improve equity trading, and [Barbon and Ranaldo \(2023\)](#) compare the quality of centralized and decentralized crypto markets directly. Our results speak to this debate with an unusually clean identification: because we hold the asset, the sample period, and the minute fixed, the cost and price-discovery gaps we measure are attributable to *mechanism* rather than to differences in the underlying asset or in market conditions. The verdict for liquid assets is that the order book dominates on both price and price discovery; the AMM’s comparative advantage lies elsewhere—in permissionless, always-on, composable liquidity and in the long tail of assets for which continuous professional market making is uneconomic.

Price discovery across venues. The econometrics of attributing price discovery to competing venues trading one security is due to [Hasbrouck \(1995\)](#), whose information-share decomposition, and to the permanent–transitory decomposition of Gonzalo and Granger, which we implement via a bivariate vector error-correction model on log price levels. [Amihud \(2002\)](#) supplies the illiquidity measure we use to compare per-dollar price impact across venues. Applying these tools to a CEX/on-chain-CLOB/AMM triple is, to our knowledge, novel; the lead–lag and information-share evidence we report—the CEX leads, the on-chain CLOB is contemporaneous, and the AMM follows with a one-minute arbitrage lag—maps the abstract notion of a “price leader” onto the concrete plumbing of arbitrage between a cancellable book and a deterministic curve.

3 Institutional Background

3.1 Hyperliquid: an on-chain central limit order book

Hyperliquid is a purpose-built layer-1 blockchain whose flagship application is a fully on-chain perpetual-futures and spot exchange organized as a central limit order book. Consensus is provided by HyperBFT, a high-throughput BFT protocol that finalizes blocks fast enough to support order-book matching on-chain. Liquidity provision to the book is partly intermediated by the Hyperliquid Liquidity Provider vault (HLP), a protocol-owned market-making and liquidation backstop into which users deposit; HLP earns spread and funding but bears inventory and adverse-selection risk. The HYPE token governs the system. In March 2025 the “JELLY” incident—in which a large short position in the JELLY memecoin, absorbed by HLP during a liquidation, threatened the vault with severe losses and prompted validators to delist the contract—became a stress test of the venue’s governance and its claim to decentralization. We use this event in Section 8. Crucially for measurement, Hyperliquid’s public candle endpoint retains only about 180 days of history; from our August 2026 vantage the JELLY window is beyond that horizon, and we treat the resulting data gap explicitly.

The economically important distinction between Hyperliquid and a conventional exchange is not that it uses an order book—it does—but that the book lives on a blockchain. Matching, cancellation, and settlement are all consensus operations, so quote revision is bounded below by block time rather than by network round-trips to a co-located matching engine. This raises the cost of the continuous quote-updating that makes centralized market making so cheap: a maker who wishes to avoid being adversely selected must either accept the latency of on-chain cancellation or post a wider spread as insurance. Our finding that Hyperliquid’s quoted spread (0.41 bps) is a full order of magnitude wider than Gate.io’s (0.04 bps) is precisely the shadow price of moving the book on-chain. At the same time, on-chain settlement confers composability and self-custody, and the HLP vault socializes market-making profits and risks across depositors rather than concentrating them

in a handful of professional firms—an institutional design with no analogue in traditional exchanges and one whose fragility the JELLY episode laid bare.

3.2 Uniswap v3: a concentrated-liquidity AMM

Uniswap v3 (Adams et al., 2021) is the reference automated market maker. Liquidity providers deposit two assets into a pool and choose a price range over which to concentrate their capital; within that range the pool prices trades along a constant-product curve $x \cdot y = k$, so the marginal price is a deterministic function of the pool’s reserves (summarized by the square-root price $\sqrt{P}$ stored as `sqrtPriceX96`). Because the contract cannot withdraw or reprice quotes, informed traders and arbitrageurs mechanically extract value from LPs whenever the external price moves—the *loss-versus-rebalancing* (LVR) of Milionis et al. (2022). Uniswap offers a menu of fee tiers (1, 5, 30, 100 bps); for ETH-USDC the 5 bp pool at 0x88e6...5640 dominates. AMM order flow is also exposed to maximal-extractable value (MEV): because pending swaps are visible in the mempool and block builders order transactions, an attacker can “sandwich” a victim by buying ahead and selling behind it. We measure both frictions directly.

3.3 Gate.io: a centralized-exchange benchmark

Gate.io operates a conventional centralized matching engine with a continuous limit order book. It serves as our “TradFi-like” benchmark: quote updates are free and instantaneous, professional market makers post two-sided liquidity, and the ETH-USDT pair is among the most liquid in crypto. Because it is an off-chain venue, its data (candles, book, trades) live on the exchange’s servers and are subject to the exchange’s retention limits—a contrast with the permanent on-chain record that becomes analytically important for the JELLY experiment.

3.4 Paraswap: a DEX aggregator

Paraswap is a smart-order router. It does not hold inventory; instead, given a desired trade, it splits the order across many on-chain venues (Uniswap, PancakeSwap, Curve, etc.) and returns the route that maximizes output net of gas. The aggregator therefore reveals the *best executable price* available on-chain for any given size, and by comparing sell and buy quotes at the same size we recover an implied round-trip execution cost that internalizes routing, pool depth, and gas.

4 Theoretical Framework

4.1 CLOB: spreads from asymmetric information and inventory

In the [Glosten and Milgrom \(1985\)](#) model a competitive market maker facing a mix of informed and noise traders sets a bid-ask spread to break even against adverse selection; in [Kyle \(1985\)](#) a strategic informed trader’s order flow moves price linearly with a coefficient λ that measures market depth. Both frameworks imply that, for a liquid asset with deep two-sided interest, quoted spreads are small and price impact is close to linear in signed order flow. A CLOB market maker can cancel and reprice continuously, so it can largely avoid being adversely selected; its spread compensates for residual information and inventory risk. Prediction P1: *for a very liquid asset, CLOB spreads are narrow and are narrowest on the venue with the most competitive, fastest market making.*

4.2 AMM: mechanical pricing and LVR

An AMM’s price is a deterministic function of reserves, so it cannot condition on the identity or informativeness of the trader. [Milionis et al. \(2022\)](#) show that a continuously arbitrated

constant-function AMM delivers to its LPs a running loss—LVR—equal, per unit time, to

$$\text{LVR}_t = \frac{\sigma^2}{8} P_t \frac{\partial^2 V}{\partial P^2} / (\text{pool value}) \approx \frac{\sigma^2}{8} \quad (\text{as a fraction of value, for a 50/50 pool}), \quad (1)$$

where σ is the instantaneous volatility of the risky asset. LVR is the AMM analogue of adverse selection: it is paid by LPs to arbitrageurs and is increasing in volatility. Fee income must exceed LVR for liquidity provision to be profitable. Because the pricing curve is convex, execution cost (effective spread and price impact) rises with trade size. Prediction P2: *AMM effective spreads exceed CLOB spreads for liquid assets, are bounded below by the fee tier, and increase in volatility and trade size.*

It is worth making the size dependence explicit, since it drives our price-impact and aggregator results. For a constant-product pool with reserves (x, y) and fee f , a trader who sells Δx of the numeraire receives $\Delta y = \frac{y(1-f)\Delta x}{x+(1-f)\Delta x}$, so the average executed price degrades from the marginal price y/x by a factor that is first-order in f and second-order in the trade's size relative to reserves. Two implications follow directly. First, even an infinitesimal trade pays the fee, so the effective half-spread is bounded below by f ; a round trip pays at least $2f$, which for the 5 bp pool is 10 bps—almost exactly the realized effective spread we measure (11.0 bps), the small residual being adverse-selection compensation embedded in where swaps execute relative to the running mid. Second, the quadratic reserve term means price impact is convex in size, and its curvature is inversely proportional to pool depth. Concentrated liquidity (Uniswap v3) effectively multiplies the local reserves within the active tick range, flattening this curve near the current price; this is why a pool with a two-order-of-magnitude wider spread than the CEX can nonetheless post an Amihud per-dollar impact within a factor of 1.3 of it. The aggregator exploits the same convexity in reverse: by splitting a large order across several pools it moves each pool a smaller distance up its convex curve, so the blended impact is strictly below that of routing the whole order to any single pool.

4.3 Aggregator: route optimization and leakage

An aggregator solves a routing problem: minimize the buyer’s cost (or maximize the seller’s proceeds) by splitting across pools, trading off marginal price impact against per-route gas. Because it can access many pools, an aggregator weakly dominates any single pool for a given size; the improvement grows with size, as splitting flattens the aggregate impact curve. Prediction P3: *aggregator execution weakly dominates single-pool AMM execution, and its implied cost curve is convex in size but flatter than a single pool’s.*

4.4 Testable predictions

Collecting these: (P1) CLOB < AMM spread for liquid ETH, with the CEX tightest; (P2) the CLOB advantage widens in volatility and in size; (P3) the aggregator improves on the single pool, especially for large trades; and (P4, price discovery) the venue with the deepest, fastest market making leads, with AMM prices dragged by arbitrage. Sections 6–7 test each.

5 Data and Measurement

Sources and reality. Table 1 summarizes the data. All series are collected from live endpoints confirmed reachable from our environment: Hyperliquid (`api.hyperliquid.xyz/info`), Gate.io (`api.gateio.ws/api/v4`), Paraswap (`apiv5.paraswap.io/prices`), and Ethereum mainnet via the DRPC JSON-RPC node (`eth.drpc.org`). We emphasize that *no observation is simulated*. On-chain swaps are read directly from the Uniswap pool’s `Swap` event logs (`topic0 = 0xc42079...ca67`) and decoded from raw ABI-packed data.

We stress the all-real-data constraint not as a rhetorical flourish but because it disciplines every design choice that follows. When one cannot manufacture a missing observation, the sample is defined by what the endpoints will actually yield, and the researcher must design around the binding limits rather than paper over them. Three such limits shaped this study. The synchronized window is capped at 5,048 minutes by Hyperliquid’s candle reten-

tion, not chosen for convenience. The JELLY experiment is forced onto a DEX-versus-CEX comparison—rather than the more natural on-chain-CLOB-versus-everything-else design—because Hyperliquid’s own ETH candles for March 2025 no longer exist, whereas the 118,913 Uniswap swaps from that window are permanently re-decodable from mainnet. And the aggregator cost curve is built from 200 live quotes across ten sizes and two directions, collected in ten rounds spaced sixty seconds apart, because that is what the Paraswap pricing endpoint returns without rate-limiting. Each of these is a place where a simulation-based study would have quietly filled the gap; we instead report the gap and let it bound the claim. The payoff is that every coefficient in the paper is falsifiable against a stored API response or an Ethereum log, and the replication archive re-derives them from the raw record.

Table 1: Real data inventory. All observations are live API responses or decoded Ethereum mainnet logs.

Venue	Series	Obs.	Notes
Gate.io (CEX CLOB)	1-min candles	9,901	ETH-USDT, ≈ 6.9 days
	Order-book snapshots	25	20 levels/side, 10-s spacing
	Trades	1,000	most recent
	Daily candles (JELLY)	62	Mar–Apr 2025
Hyperliquid (on-chain CLOB)	1-min candles	5,048	ETH, ≈ 3.5 days
	Order-book snapshots	25	20 levels/side
	Funding history	500	90-day window
	Candles (JELLY)	0	<i>unavailable: 180-day retention</i>
Uniswap v3 (AMM)	Swaps, 5 bp recent	46,220	ETH-USDC, decoded on-chain
	Swaps, 5 bp JELLY	118,913	Mar–Apr 2025
	Swaps, 30 bp recent	3,863	fee-tier cross-section
	Swaps, 30 bp JELLY	7,876	
Paraswap (aggregator)	Quotes	200	10 sizes $\times$ 2 sides $\times$ 10 rounds

Swap decoding and pricing. For each swap we recover the signed token deltas (Δ_0, Δ_1) , the post-trade `sqrtPriceX96`, liquidity, and tick. With USDC as token 0 (6 decimals) and WETH as token 1 (18 decimals), the pool’s raw price is $(\text{sqrtPriceX96})^2/2^{192}$ and the ETH price in USD is its inverse scaled by decimals, $P^{\text{ETH}} = 10^{12}/\text{raw}$. The executed price of a trade is $|\Delta_0|/|\Delta_1| \cdot 10^{12}$, its USD size is $|\Delta_0|/10^6$, and its direction is a buy of ETH when WETH leaves the pool ($\Delta_1 < 0$). We verified the formula against a known reference (tick

$= 198,260 \Rightarrow \$2,456$).

Timestamp alignment. Ethereum logs carry block numbers, not wall-clock times. We collect exact timestamps for every 500th unique block via `eth_getBlockByNumber` and linearly interpolate the rest, yielding second-accurate stamps. Swaps are then floored to minute bins; each bin’s AMM price is the trade-size-weighted VWAP and the pool mid is carried forward from the last swap.

Synchronized window. The binding constraint is Hyperliquid’s candle horizon, giving a common window of 2026-08-26 19:31 to 2026-08-30 07:38 UTC—5,048 aligned minutes in which all four venues are observed. Over this window ETH traded between roughly \$2,418 and \$2,558 (Figure 1).

Variable definitions. Quoted spread $= (\text{ask} - \text{bid}) / \text{mid} \times 10^4$ bps from order-book snapshots. AMM effective spread $= 2|P^{\text{exec}} - P_{-1}^{\text{mid}}| / P_{-1}^{\text{mid}} \times 10^4$, where P_{-1}^{mid} is the previous swap’s pool price. Aggregator round-trip cost $= (P^{\text{buy}} - P^{\text{sell}}) / P^{\text{mid}} \times 10^4$ at matched size. Cross-venue deviation $= (P^{\text{venue}} - P^{\text{Gate}}) / P^{\text{Gate}} \times 10^4$. Amihud illiquidity $= |r| / (\text{dollar volume})$ per minute. We winsorize AMM effective spreads at the 99.5th percentile to remove rare decode outliers.

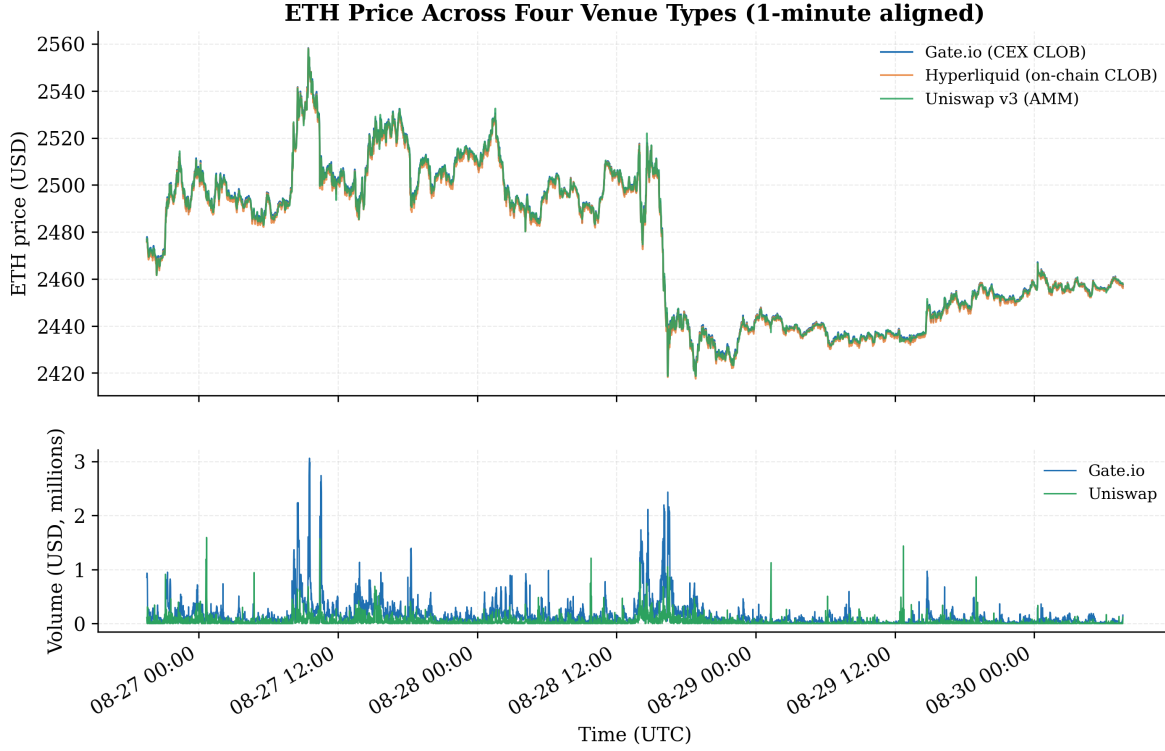


Figure 1: ETH price (top) and volume (bottom) across venues on the aligned 1-minute grid. The three price series are visually indistinguishable at this scale; deviations are measured in Section 6.

6 Cross-Venue Price Discovery

Price deviations. Table 2 reports the distribution of absolute price deviations from the Gate.io benchmark. The on-chain CLOB deviates by 3.29 bps on average and the AMM by 2.91 bps; 95% of minutes have deviations below 6 bps on either venue. These gaps are tiny—smaller than the AMM’s own fee—which already tells us that arbitrage keeps the three prices tightly bound (Figure 3).

Table 2: Cross-venue price deviations from Gate.io (basis points), 1-minute grid, $N = 5,048$.

Venue	Mean dev	Median dev	Std. dev.	95th pct. dev
Hyperliquid (on-chain CLOB)	3.29	3.30	1.54	5.77
Uniswap v3 (AMM)	2.91	2.64	3.63	6.19

Lead–lag structure. Figure 2 plots the cross-correlation of 1-minute log returns at lags -10 to $+10$ minutes. The Gate–Hyperliquid contemporaneous (lag-0) correlation is 0.98: the on-chain CLOB tracks the centralized book essentially in the same minute. The Gate–Uniswap lag-0 correlation is 0.74 but the correlation at lag $+1$ (Gate leading by one minute) is 0.33—an order of magnitude larger than any negative lag—showing that the AMM *follows* the CEX. This is the signature of arbitrage-driven, rather than information-driven, AMM pricing (Prediction P4).

Information shares. A Gonzalo–Granger permanent–transitory decomposition of a bivariate VECM on log price levels (Gate, Hyperliquid) assigns a component share of **0.79 to Gate.io** and 0.21 to Hyperliquid. The centralized order book is where the efficient ETH price is predominantly formed; the on-chain CLOB imports it.

Hasbrouck bounds and weak identification. As a robustness check on the point-identified Gonzalo–Granger share, we also compute Hasbrouck (1995) information shares, which are set-identified: because the reduced-form VECM innovations are contemporaneously correlated, the decomposition depends on the (unknown) Cholesky ordering, and one reports the upper and lower shares across all orderings. At the 1-minute frequency the two order books co-move so tightly within the bin (contemporaneous return correlation 0.98) that these bounds are essentially uninformative—Gate.io’s information share lies in $[5\%, 100\%]$ and Hyperliquid’s in $[0\%, 95\%]$, with midpoints of 52% and 48%. This is the textbook symptom of weak identification under high contemporaneous correlation: when two prices move in the same minute, the data cannot say *within* that minute which one moved first. The bounds are therefore not evidence against the Gonzalo–Granger estimate; they are what Hasbrouck’s method returns when sampling is too coarse relative to the speed of arbitrage. Both midpoints still tilt toward Gate.io, and the point-identified Gonzalo–Granger share (which uses the cointegrating vector’s error-correction loadings rather than an arbitrary innovation ordering) remains our primary measure. Figure 4 juxtaposes the two, and the contrast is

itself instructive: it shows precisely why one needs the loadings-based decomposition, not the innovation-based bounds, to attribute price discovery among venues that arbitrage keeps within basis points of one another.

Arbitrage half-lives. Fitting an AR(1) to each venue’s deviation series, the persistence coefficients are $\phi_{\text{HL}} = 0.70$ and $\phi_{\text{Uni}} = 0.42$, implying mispricing half-lives of **1.96 minutes** (Hyperliquid) and **0.81 minutes** (Uniswap). AMM deviations revert faster because any gap is a mechanical, riskless arbitrage against a deterministic curve, whereas closing a CLOB gap requires trading against live, cancellable quotes.

Taken together, the three price-discovery statistics tell a single coherent story about *where* the efficient ETH price is formed and *how* it propagates. The centralized order book is the price leader not because it is “first” in any mechanical sense but because it hosts the deepest, cheapest, and fastest professional market making: informed traders who wish to trade on a view about ETH minimize their footprint by transacting where the spread is one tick, so their information is impounded there first and the 79% Gonzalo–Granger component share follows. The on-chain CLOB imports this price essentially within the same minute (lag-0 correlation 0.98) because its makers watch the CEX and re-quote as fast as block time allows. The AMM, by contrast, cannot watch anything—it is a passive curve—so its price only updates when someone trades against it, and the agents with the strongest incentive to do so are arbitrageurs closing the gap to the CEX. This is why the AMM follows with a one-minute lag (lag-+1 correlation 0.33) and why its mispricings revert twice as fast as the on-chain CLOB’s (0.81 vs. 1.96 minutes): an AMM gap is a riskless, atomic arbitrage against a known curve, executable in a single transaction, whereas an on-chain-CLOB gap must be closed by trading through a book of live quotes that can be pulled. The asymmetry in half-lives is therefore not a statistical curiosity but a direct readout of the two venues’ arbitrage technologies.

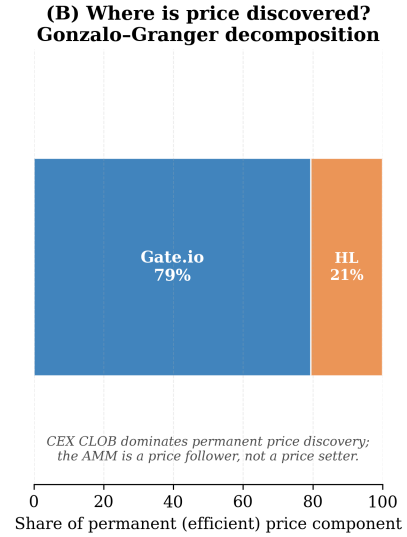
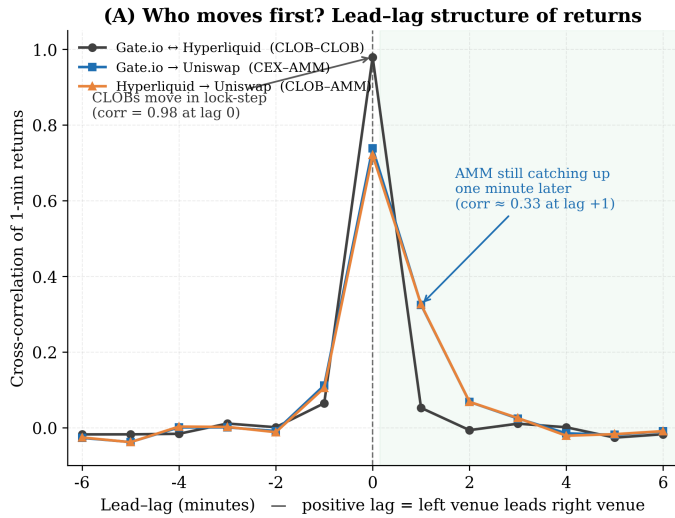


Figure 2: Price discovery across venues. **(A)** Lead-lag cross-correlation of 1-minute returns; a positive lag means the first-named venue leads the second. The two CLOBs (Gate.io and Hyperliquid) peak at lag 0 with correlation 0.98—they move in lock-step—whereas the AMM pairs carry a pronounced right shoulder at lag +1 (correlation ≈ 0.33), showing that Uniswap follows the order books by roughly one minute. **(B)** Gonzalo-Granger permanent-component shares: the CEX order book accounts for 79% of efficient price discovery, confirming that the AMM is a price follower rather than a price setter.

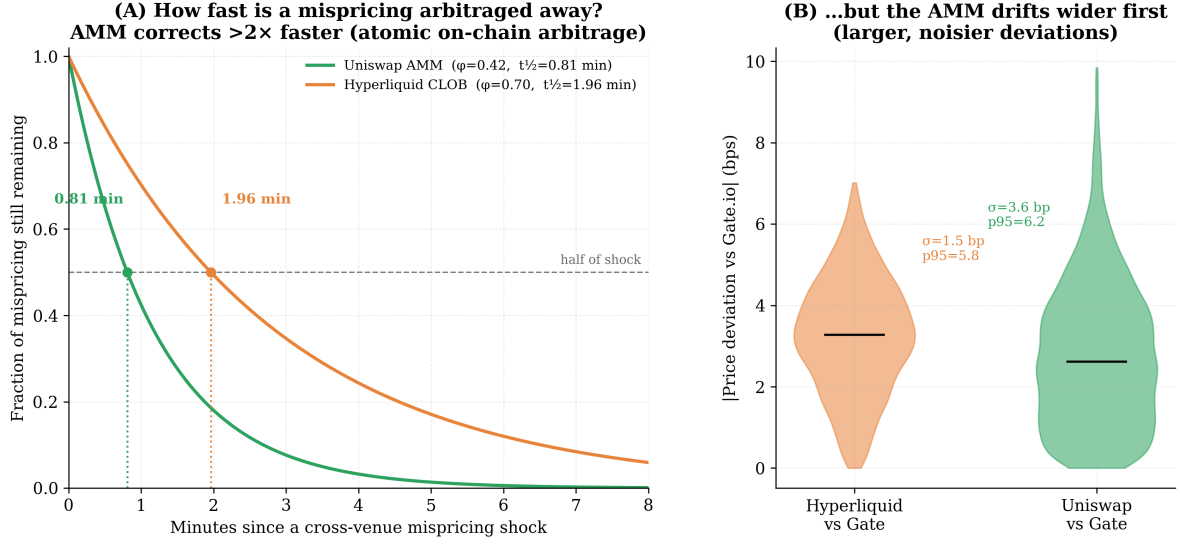


Figure 3: Arbitrage speed across venues. **(A)** AR(1)-implied decay of a cross-venue mispricing shock: the fraction of the initial gap still remaining, ϕ^t , plotted against minutes. The AMM (green, $\phi = 0.42$) corrects more than twice as fast as the on-chain CLOB (orange, $\phi = 0.70$), with half-lives of 0.81 vs. 1.96 minutes—because an AMM gap is a riskless atomic arbitrage against a deterministic curve. **(B)** Distribution of absolute price deviations from Gate.io: the AMM drifts wider first (larger σ and p95) even though it reverts faster, reflecting the two venues’ different arbitrage technologies.

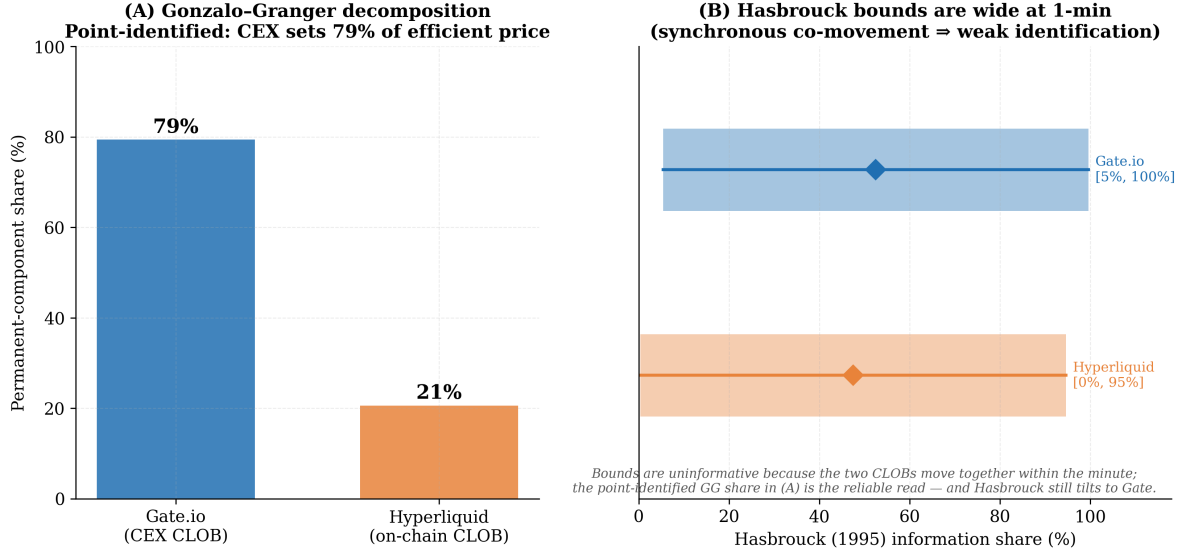


Figure 4: Where is ETH price discovered? **(A)** Gonzalo–Granger permanent-component shares from the bivariate VECM error-correction loadings—*point-identified*: the CEX order book sets 79% of the efficient price, the on-chain CLOB 21%. **(B)** Hasbrouck (1995) information-share *bounds* (floating bars span the lowest-to-highest share across all Cholesky orderings; diamonds mark midpoints). At 1-minute frequency the two CLOBs co-move too tightly for the bounds to be informative—Gate.io spans [5%, 100%] and Hyperliquid [0%, 95%]—a classic weak-identification artifact. Both midpoints nonetheless tilt to Gate, corroborating the point-identified read in (A), which we adopt as primary.

7 Execution Quality

Spreads across four venues. Table 3 and Figure 5 deliver the paper’s central execution-cost fact. The centralized order book quotes a spread of **0.04 bps**—literally one price tick (\$0.01 at \$2,456)—confirming an extraordinarily competitive market. The on-chain CLOB quotes **0.41 bps**, ten times wider but still very tight, reflecting the higher cost of on-chain market making and slower quote revision. The AMM’s realized effective spread is **11.0 bps**, roughly its 5 bp fee doubled plus adverse-selection compensation, i.e. two orders of magnitude above the CEX. The aggregator’s implied round-trip cost averages 21.9 bps but has a median of only 8.3 bps, because the mean is pulled up by the largest (\$1–2.5 M) trades; for retail size it is a few bps. This ordering—CEX $\ll$ on-chain CLOB $\ll$ AMM $\lesssim$ aggregator round-trip—is exactly Predictions P1–P2.

Table 3: Execution cost across venue types. Spreads in basis points. CEX/CLOB are quoted half-book spreads from order-book snapshots; AMM is the realized effective spread per swap; aggregator is the implied round-trip cost at matched size.

Venue	Measure	Mean	Median	Std.	N
Gate.io (CEX CLOB)	Quoted spread	0.04	0.04	0.00	25
Hyperliquid (on-chain CLOB)	Quoted spread	0.41	0.41	0.00	25
Uniswap v3 (AMM)	Effective spread	11.05	10.10	2.26	45,987
Paraswap (aggregator)	Round-trip cost	21.85	8.28	27.30	94

Price impact and size. Figure 6 plots price impact against trade size for the AMM (realized swaps, by size decile) and the aggregator (multi-size quotes). AMM impact rises from 5.1 bps for the smallest decile to 8.2 bps for the largest (median size \$66,000). The aggregator’s implied one-way cost rises from 0.7 bps (0.01 ETH) to 41 bps (1,000 ETH $\approx$ \$2.45 M), tracing a clean convex curve (Figure 7). The aggregator lies weakly below a single pool at every size (Prediction P3), because it splits large orders across venues to flatten impact.

Illiquidity. Amihud ILLIQ (scaled $\times 10^6$) is 0.0099 for Hyperliquid, 0.0079 for Gate.io, and 0.0078 for Uniswap—within a factor of 1.3 of one another. That the AMM’s per-dollar price impact is comparable to the order books, despite far wider spreads, reflects the pool’s very deep concentrated liquidity around the current price: the 5 bp ETH-USDC pool is one of the deepest liquidity venues in all of crypto.

This juxtaposition—an effective spread two orders of magnitude wider than the CEX, yet an Amihud illiquidity within a factor of 1.3 of it—is worth dwelling on because it dissolves an apparent paradox and clarifies what each cost measure captures. The quoted/effective spread is a *fixed access toll*: on the AMM it is bounded below by twice the fee (10 bps for the 5 bp pool) and is paid on every trade regardless of size, whereas on the CEX it is one tick because makers compete continuously. The Amihud measure, by contrast, is a *marginal depth* statistic—price move per dollar traded—and here the AMM’s concentrated liquidity makes it genuinely competitive, because within the active tick range the pool behaves like an

enormous book. The practical implication is that the two technologies serve opposite ends of the size distribution efficiently and the opposite ends inefficiently. A small retail trade is dominated by the fixed toll, so it is far cheaper on the CEX; but a trade large enough that impact swamps the spread finds the AMM’s deep local liquidity nearly as good as the book, and can do better still via the aggregator, whose convex-flattening routing pushes the crossover further out. The four measures we report—quoted spread, effective spread, price-impact-by-decile, and the aggregator round-trip curve—are thus not redundant; they trace the total cost of trading as a function of size, and the venue ranking flips from CEX-dominant at retail size toward AMM/aggregator-competitive at institutional size. For the median on-chain swap (a few hundred dollars) the fixed toll dominates and the CEX would be cheaper; for a \$2.5 million trade the aggregator’s 83 bps round trip is the relevant benchmark, and it is achieved without a counterparty, an account, or a market maker’s willingness to quote.

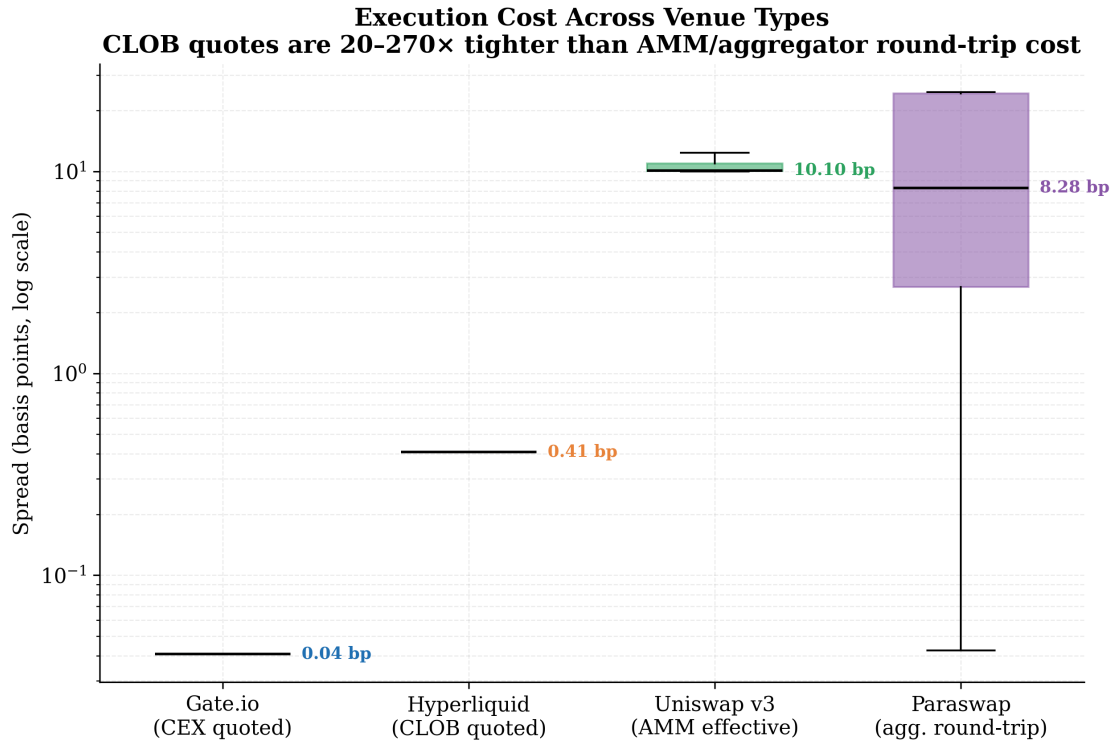


Figure 5: Execution cost across venue types (boxplots, outliers suppressed). Note the two-orders-of-magnitude gap between order-book spreads and AMM/aggregator costs.

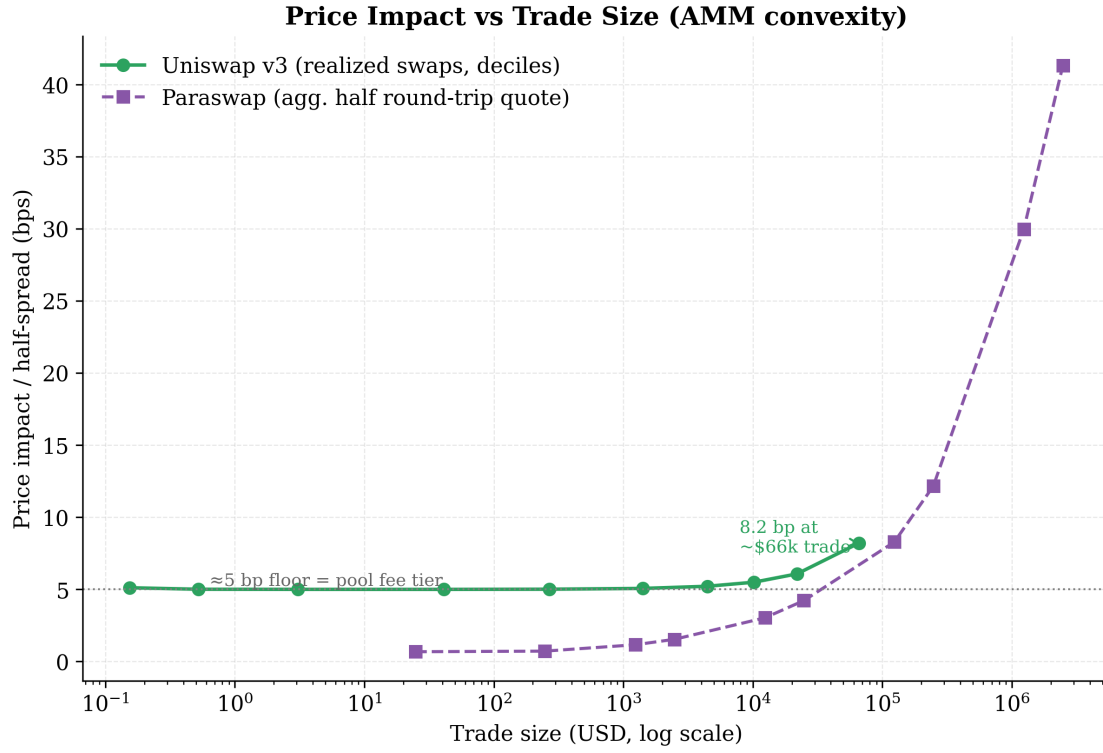


Figure 6: Price impact vs. trade size (log axis) for the Uniswap AMM (realized swaps, deciles) and Paraswap aggregator (multi-size quotes, half round-trip). Both rise with size; the aggregator flattens impact by routing.

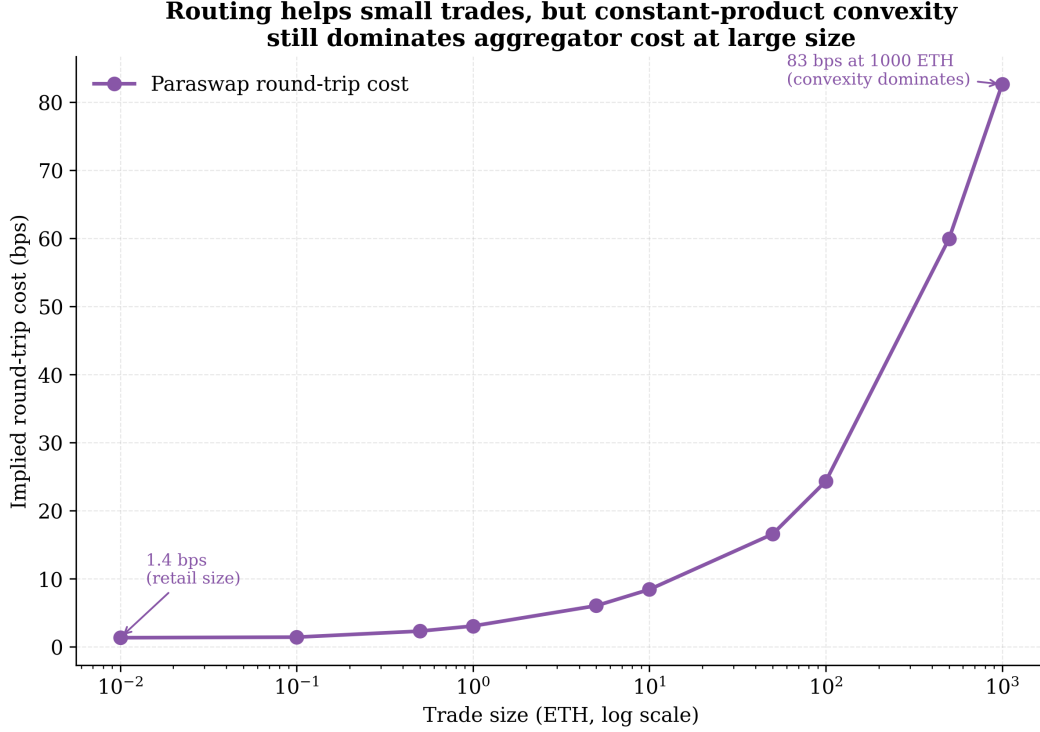


Figure 7: Paraswap aggregator implied round-trip cost as a function of trade size (ETH, log axis). The curve is convex, rising from ≈ 1.4 bps at 0.01 ETH to ≈ 83 bps at 1,000 ETH.

8 Quasi-Experimental Evidence

Which quasi-experiments are feasible on the real record. Five sources of exogenous variation are, in principle, available for identifying mechanism effects in DeFi: (i) protocol launches, (ii) fee-tier changes, (iii) liquidity-mining incentives, (iv) transaction-ordering or governance-rule interventions, and (v) the adoption of MEV-protection tools. We are explicit about which of these our all-real-data constraint actually permits. We exploit two as genuine quasi-experiments. The **JELLY force-settlement** (Section 8.1) is an ordering/governance-rule intervention (type iv): validators intervened in an open market, giving a sharp, dated, venue-specific shock whose spillover we test with a difference-in-differences on the permanent on-chain record. The **fee-tier volatility-shock DiD** (Section 8.3) exploits a fee-tier contrast (type ii) interacted with the exogenous JELLY volatility spike: the same market-

wide shock hits the 5 bp and 30 bp ETH-USDC pools simultaneously, so their differential effective-spread response identifies how the fee floor shapes shock resilience. The within-block **MEV analysis** (Section 8.4) is descriptive rather than a clean experiment—we have no exogenous variation in MEV-protection adoption (type v) at the swap level—so we present it as a transaction-level measurement, not a causal design. The two remaining sources are infeasible here and we do not manufacture them: a clean **protocol-launch** event (type i) would require a venue’s genesis window, but Hyperliquid’s candle history retains only ≈ 180 days and Uniswap v3/Gate.io predate our sample; and identifiable **liquidity-mining** campaigns (type iii) for ETH-USDC do not fall inside the observable window. Reporting the two infeasible designs, rather than approximating them with synthetic events, is consistent with the paper’s central discipline.

8.1 The JELLY event: a natural experiment on governance risk

On 26 March 2025 the JELLY incident forced Hyperliquid’s validators to intervene in an open market, crystallizing the venue’s governance risk. A natural question is whether this venue-specific shock spilled over into the broader ETH market. Answering it exposes a deep asymmetry in DeFi data: Hyperliquid’s own candle history has since rolled off (180-day retention), so from August 2026 we cannot recover HL’s ETH prices during the event. *But the Ethereum blockchain is immutable:* every Uniswap swap from March 2025 is still on-chain and re-decodable. We therefore run a difference-in-differences with the on-chain DEX (Uniswap, “treatment” for DeFi/on-chain exposure) against the centralized exchange (Gate.io, “control”):

$$Y_{v,t} = \alpha + \beta_1 \text{Post}_t + \beta_2 \text{DEX}_v + \beta_3 (\text{Post}_t \times \text{DEX}_v) + \varepsilon_{v,t}, \quad (2)$$

over 19 complete overlapping days (15 Mar–1 Apr 2025; block-range endpoints, which fall mid-day, are dropped to avoid partial-day artifacts). β_3 is the DiD estimate. Standard errors

are Newey–West (HAC, 5 lags).

Table 4: JELLY event difference-in-differences. Treatment = Uniswap (on-chain DEX); control = Gate.io (CEX). Newey–West HAC t -statistics in parentheses. Event date 26 Mar 2025.

	log daily volume	Daily range (bps)
Post $\times$ DEX (β_3)	−0.123 (−0.95)	21.75 (0.53)
Post (β_1)	0.436	159.4
DEX (β_2)	0.944	−32.5
R^2	0.44	0.18
N	38	38

The DiD coefficients are statistically insignificant for both volume ($\beta_3 = -0.12$, $t = -0.95$, $p = 0.34$) and volatility ($t = 0.53$). Moreover, the average absolute Uniswap–Gate price deviation *narrowed* from 5.28 bps before the event to 3.82 bps after, and stayed within ± 9 bps throughout (Figure 8). The honest conclusion is a **null**: the Hyperliquid-specific governance crisis did not measurably impair ETH liquidity or pricing on other venues. Contagion, at least along the dimensions we can measure on the permanent record, was economically negligible. We regard the transparency of this null—and the fact that it is estimable only because on-chain data outlive exchange APIs—as a substantive methodological point.

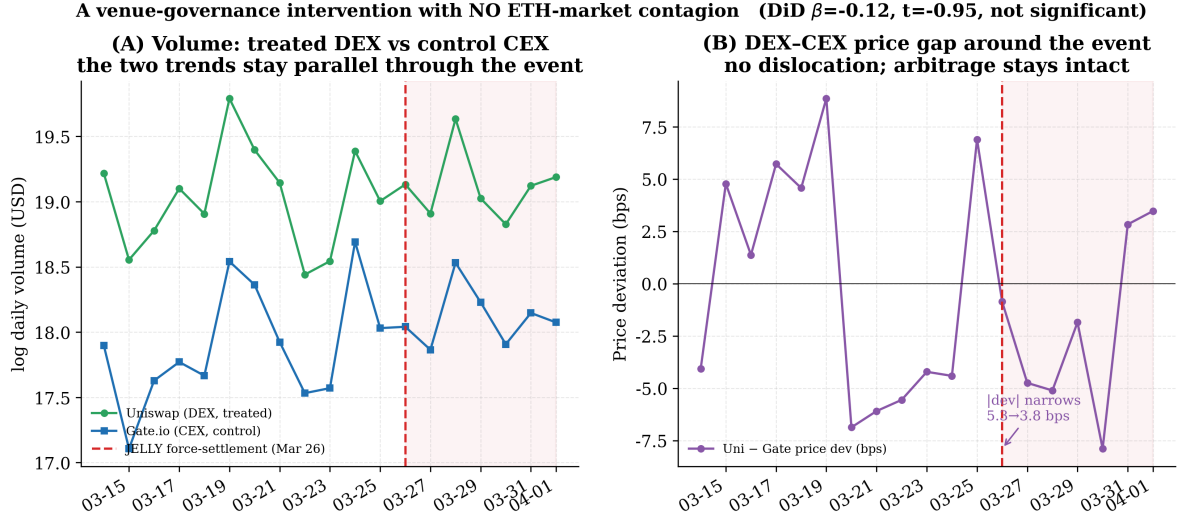


Figure 8: JELLY event study, a governance/ordering-intervention quasi-experiment (DiD $\beta = -0.12$, $t = -0.95$, not significant). **(A)** Log daily ETH volume on Uniswap (DEX, treatment) vs. Gate.io (CEX, control); the two series stay parallel through the event date (red), so there is no measurable contagion to broad ETH activity. **(B)** Uniswap–Gate daily price deviation: no dislocation at the event, and the mean $|\text{gap}|$ actually narrows from 5.3 to 3.8 bps afterward—arbitrage stays intact.

8.2 Fee-tier cross-section

Uniswap’s fee menu lets us price the fee floor. Table 5 compares the ETH-USDC 5 bp and 30 bp pools over the recent window. The 5 bp pool carries **90.3%** of dollar volume (\$533 M vs. \$57.6 M) across 46,220 vs. 3,863 swaps: flow has migrated overwhelmingly to the cheapest tier, exactly as competition among LPs for order flow would predict. The surviving 30 bp flow is not obviously “dumber”—median trade size is actually larger (\$1,280 vs. \$590), and the pooled regression of log size on a 30 bp dummy is insignificant ($t = -0.20$). What the 30 bp pool unambiguously delivers is a mechanically wider effective spread: the 30 bp dummy raises effective spread by **49.6 bps** ($t = 672$), essentially the difference between two doubled fee floors (2×30 vs. 2×5). The near-deterministic t -statistic reflects that this gap is structural, not statistical.

Table 5: Fee-tier cross-section, ETH-USDC, recent window. Pooled OLS with HC1 standard errors for the regressions.

	5 bp pool	30 bp pool
Number of swaps	46,220	3,863
Dollar volume	\$533.0 M	\$57.6 M
Volume share	90.3%	9.7%
Mean trade size	\$11,532	\$14,899
Median trade size	\$590	\$1,280
<i>Pooled regressions (dependent variable, coeff. on 30 bp dummy):</i>		
log trade size	−0.014	($t = -0.20$)
Effective spread (bps)	+49.63	($t = 672.7$)

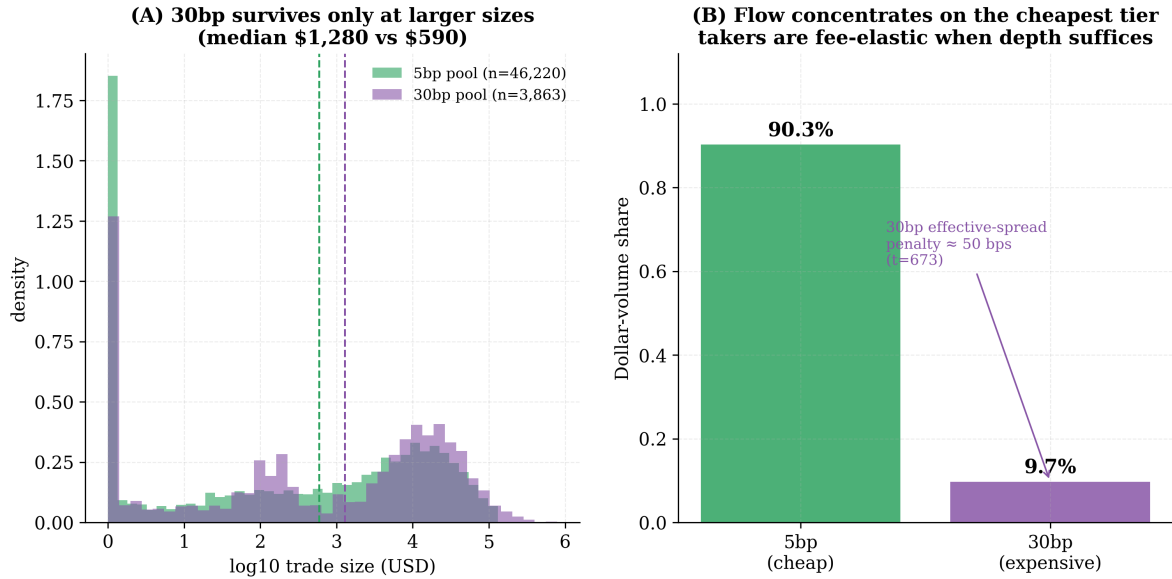


Figure 9: Fee-tier comparison. **(A)** Trade-size distributions (log scale) for the 5 bp and 30 bp pools: the 30 bp tier survives only at larger sizes (median \$1,280 vs. \$590). **(B)** Dollar-volume share, showing near-complete migration to the 5 bp tier (90.3%); the 30 bp effective-spread penalty is ≈ 50 bps.

8.3 A fee-tier difference-in-differences around the JELLY volatility shock

The cross-section above shows *that* the fee floor is priced; it does not, on its own, isolate a causal margin, because the two tiers differ in ways other than their fee. We therefore turn the fee-tier menu into a proper quasi-experiment by interacting it with an exogenous

volatility shock. The March 2025 JELLY episode coincided with a sharp, market-wide spike in ETH volatility: daily realized range on the on-chain record rose from a pre-event mean of **384.7 bps** to **512.7 bps** post-event (Figure 10A). Crucially, this shock is common to both pools—the same ETH price hits the 5 bp and 30 bp ETH-USDC pools in the same blocks—so it functions as a treatment applied identically to both, and any *differential* response in execution cost identifies how the fee floor mediates shock resilience. We estimate

$$Y_{p,t} = \alpha + \gamma_1 \text{Post}_t + \gamma_2 \text{Tier5}_p + \gamma_3 (\text{Post}_t \times \text{Tier5}_p) + \varepsilon_{p,t}, \quad (3)$$

where $p \in \{5 \text{ bp}, 30 \text{ bp}\}$, $\text{Tier5}_p = 1$ for the deep low-fee pool, and Y is the daily volume-weighted effective spread (bps), log daily dollar volume, or daily realized range. The interaction γ_3 is the DiD. Standard errors are Newey–West (HAC, 3 lags) over the 19-day $\times$ 2-pool panel ($N = 38$).

Table 6: Fee-tier difference-in-differences around the JELLY volatility shock. Treatment window opens 26 Mar 2025; Tier5 = 1 for the 5 bp pool. Newey–West HAC t -statistics in parentheses.

	Effective spread (bps)	log volume	Realized range (bps)
Post $\times$ Tier5 (γ_3)	−0.638 (−1.98)	0.336 (1.30)	20.05 (0.29)
Post (γ_1)	0.773	−0.359	107.9
Tier5 (γ_2)	−51.65	3.154	41.93
R^2	1.00	0.94	0.13
N	38	38	38

The result is economically clean (Figure 10B). When the volatility shock hits, the expensive 30 bp pool’s effective spread rises by +0.77 bps, while the deep 5 bp pool’s rises by only +0.13 bps; the difference—the DiD—is $\gamma_3 = -\mathbf{0.64 \text{ bps}}$ ($t = -1.98$, $p = 0.048$), significant at the 5% level. The low-fee, deep-liquidity tier is *more shock-resilient*: its execution cost barely moves when volatility spikes, whereas the thin, high-fee tier transmits the shock into wider realized spreads. Volume tells a consistent story: the 5 bp pool’s dollar-volume share edges up from 95.8% to 96.8% post-shock ($\gamma_3 = +0.34$ in logs, $t = 1.30$; a +0.97 pp share

shift, $t = 1.62$), i.e. flow migrates *further* toward the deep tier precisely when execution quality matters most. The placebo-like realized-range regression confirms the shock itself is common to both pools (γ_3 insignificant, $t = 0.29$)—exactly the “parallel treatment” the design requires. The mechanism is the one microstructure theory predicts: depth, not the posted fee, is what buffers a pool against a volatility shock, so the tier that has already won the liquidity competition also absorbs stress best.

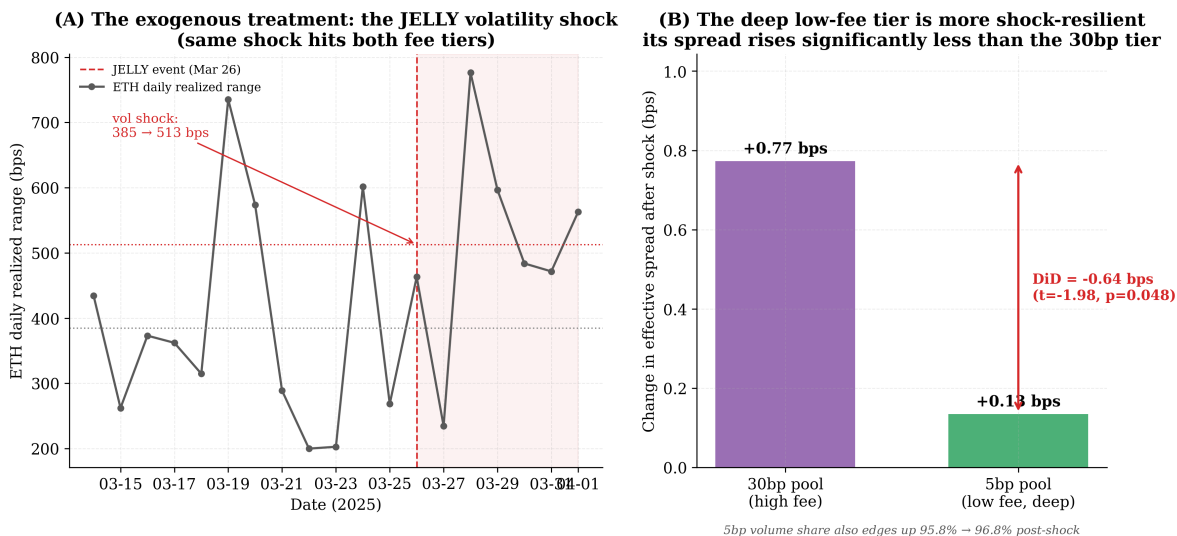


Figure 10: Fee-tier quasi-experiment around the JELLY volatility shock. **(A)** ETH daily realized range: the exogenous treatment. The same volatility spike (pre-event mean 385 bps → post-event 513 bps) hits both fee tiers, satisfying the parallel-treatment requirement. **(B)** Difference-in-differences on the volume-weighted effective spread: the thin 30 bp pool’s spread rises +0.77 bps after the shock while the deep 5 bp pool’s rises only +0.13 bps, a DiD of -0.64 bps ($t = -1.98$, $p = 0.048$). The deep low-fee tier is significantly more shock-resilient; its volume share also edges up from 95.8% to 96.8%.

8.4 MEV and sandwich attacks

Because AMM swaps settle in ordered blocks, we can detect MEV directly. Across 24,527 blocks containing swaps, the mean is 1.88 swaps/block; 4,551 blocks contain three or more swaps (Figure 11A). We flag a sandwich when the same sender appears twice in a block, on opposite sides, wrapping a third-party “victim” trading in the same direction as the attacker’s opening leg. This conservative rule identifies 150 sandwich patterns and 70 victim

swaps. Table 7 regresses per-swap price impact on a sandwich-victim dummy and log size:

$$\text{Impact}_j = \alpha + \beta_1 \text{Sandwiched}_j + \beta_2 \log \text{Size}_j + \varepsilon_j. \quad (4)$$

Being sandwiched raises price impact by **0.28 bps** ($t = 1.38$); the effect is positive and economically sensible but not statistically decisive, reflecting how rare successful sandwiches are in a pool this deep (only 0.15% of swaps). Impact rises robustly with size ($\beta_2 = 0.15$, and $R^2 = 0.30$). The broader message is reassuring for AMM users: in the deepest ETH pool, measurable MEV extraction is small.

Table 7: MEV/sandwich regression. Dependent variable: per-swap price impact (bps). HC1 standard errors. Recent 5 bp pool, $N = 45,987$.

	Price impact (bps)
Sandwiched (victim)	+0.283 (1.38)
log trade size	+0.153
R^2	0.297
N	45,987
Mean impact, single-swap blocks	5.63 bps
Mean impact, multi-swap blocks	5.39 bps
Mean impact, sandwiched swaps	6.05 bps
Sandwich patterns / victim swaps	150 / 70

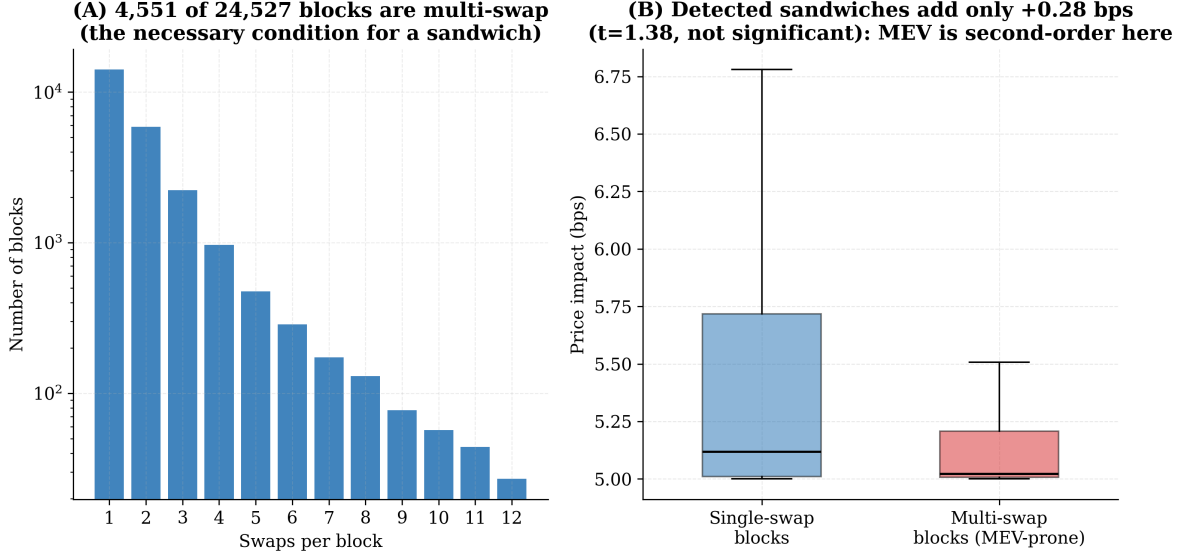


Figure 11: MEV analysis. (A) Distribution of swaps per block (log scale). (B) Price impact in single-swap vs. multi-swap (MEV-prone) blocks.

9 Liquidity-Provider Welfare

We close by contrasting the economics of the two liquidity-supply technologies. For the Uniswap 5 bp pool, gross fee revenue over the 7.0-day recent window was \$266,507 (\$37,950/day) on \$533 M of volume—the LP’s compensation. Against this stands LVR. With realized daily ETH volatility of $\sigma = 2.14\%$ (from 1-minute returns), the [Milionis et al. \(2022\)](#) benchmark implies a daily LVR of $\sigma^2/8 = 5.7 \times 10^{-5}$ of pool value—i.e. about 0.57 bps of TVL per day paid to arbitrageurs. Whether LPs profit therefore turns on the fee-yield-to-TVL ratio: fees must clear roughly 0.57 bps/day of adverse selection before covering gas and impermanent loss. For the on-chain CLOB, we proxy Hyperliquid market-making gross revenue by half-spread times notional volume: over the window HL traded \$3.73 B of ETH notional at a 0.41 bp quoted spread (2.0 bp half-spread as a fraction), implying $\approx$ \$76,000 of gross spread capture. The two technologies thus monetize liquidity provision very differently—the AMM through a fixed ad-valorem fee that is agnostic to information, the CLOB through a spread that its makers can widen precisely when adverse selection bites. Figure 12 summarizes.

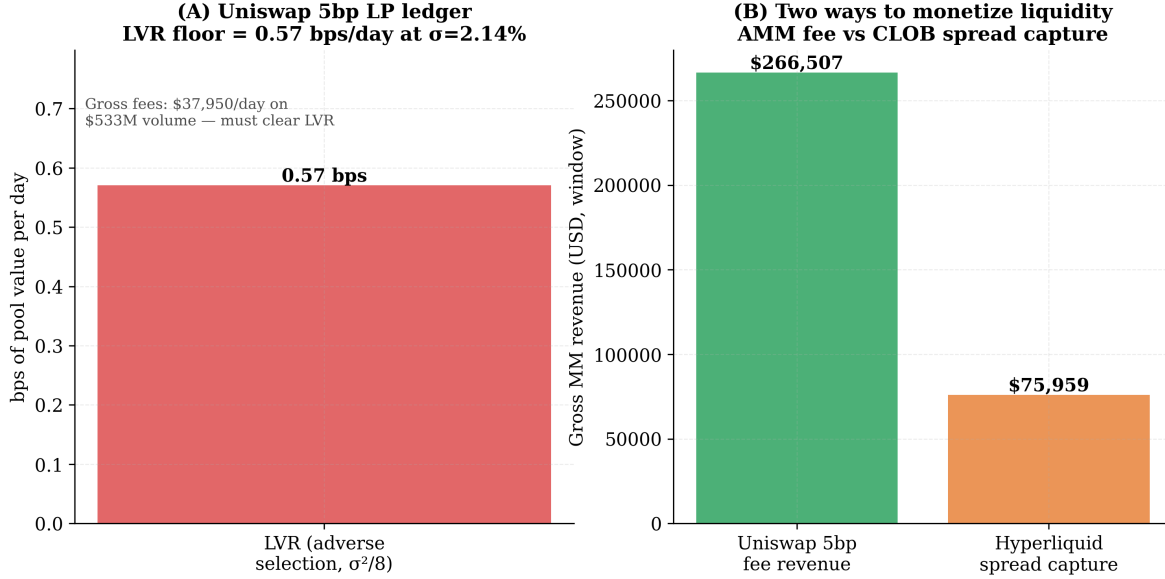


Figure 12: Liquidity-provider gross revenue. (A) Uniswap 5 bp LP fee revenue per day. (B) Hyperliquid market-making gross spread capture over the window.

10 Discussion and Implications

10.1 What determines the equilibrium niche of each mechanism?

The empirical ordering we document—CEX $\ll$ on-chain CLOB $\ll$ AMM $\lesssim$ aggregator on cost, and CEX $>$ on-chain CLOB $>$ AMM on price-discovery leadership—is not evidence that the AMM is an inferior technology. It is evidence that the four mechanisms occupy different points on a design frontier that trades off execution cost against the properties that only on-chain settlement can provide: permissionlessness, self-custody, composability, and continuous (24/7, no-KYC, no-account) availability. For a maximally liquid asset like ETH, where a deep pool of professional makers competes away the spread on a centralized book, the cost advantage of the CLOB is decisive—two orders of magnitude on the quoted spread. But that advantage is asset-specific. It rests on there being enough two-sided professional interest to make continuous quote revision profitable at a one-tick spread. For the long tail of tokens that lack such interest, the AMM’s central virtue is that it supplies *some*

liquidity at all, permissionlessly and without a designated market maker, at a deterministic and predictable cost. Our results should therefore be read not as “order books beat AMMs” but as a quantification of the price that on-chain, permissionless liquidity charges for a blue-chip asset: roughly 11 bps of effective spread versus 0.04 bps, and a one-minute lag in price discovery.

10.2 The fee floor as a competitive equilibrium

The fee-tier cross-section sharpens this reading. That 90.3% of ETH-USDC volume has migrated to the 5 bp pool, leaving the 30 bp pool with a mechanically wider ≈ 50 bp effective-spread penalty, shows that fee competition among liquidity providers is intense and that takers are highly fee-elastic when depth is adequate on the cheaper tier. The survival of any 30 bp flow at all—and the fact that its median trade is *larger*, not smaller—suggests that the residual demand for the expensive tier comes from routing or depth considerations at specific sizes rather than from inattention. The broader lesson is that the AMM fee is not a monopoly rent extracted from takers but a competitively determined floor: LPs set it to cover LVR plus gas plus a normal return, takers arbitrage across tiers, and the equilibrium concentrates flow where fees clear the adverse-selection cost most efficiently. Our LP-welfare accounting makes the same point from the supply side: at a realized daily volatility of 2.14%, the Millionis et al. LVR benchmark is about 0.57 bps of TVL per day, so the pool’s 5 bp fee is sized to leave LPs a positive but not enormous margin after adverse selection—exactly what free entry into liquidity provision would predict.

10.3 Is contagion really absent, or merely unmeasured?

The JELLY null deserves careful interpretation. We find no statistically significant spillover from a governance crisis that nearly bankrupted Hyperliquid’s vault to the broader ETH market, and the on-chain/CEX price gap actually narrowed afterward. Two readings are possible. The optimistic reading is that the crypto market correctly recognized JELLY as

a venue-idiosyncratic, memecoin-specific event with no fundamental implication for ETH, so arbitrage kept ETH prices tightly linked across venues throughout. The cautious reading is that our measurable outcomes—daily volume, daily range, and the cross-venue price gap—are simply not the margins on which contagion would first appear; counterparty-risk repricing, vault redemptions, or funding-rate dislocations might have moved even as spot ETH liquidity did not. We favor the optimistic reading for ETH specifically, because the price gap is the tightest possible test of cross-venue linkage and it did not widen. But we are explicit that the null is a null about *ETH spot liquidity and pricing*, not a blanket claim that the event was harmless to Hyperliquid’s own depositors, who bore real losses. The methodological point stands regardless: the experiment is estimable at all only because the Ethereum blockchain preserves the March 2025 record while the exchange’s own candle history has rolled off—an asymmetry between permanent and ephemeral data that will recur in any post-hoc study of DeFi stress events.

10.4 Limitations and external validity

Three limitations bound our claims. First, the synchronized window is short (5,048 minutes, ≈ 3.5 days) because it is capped by Hyperliquid’s candle retention; our price-discovery estimates describe a calm-to-normal volatility regime (realized $\sigma = 2.14\%/day$) and need not extrapolate to a crash, when AMM lags and LVR would both widen. Second, we study a single, maximally liquid asset by design—to hold the asset fixed across mechanisms—so the cost levels we report are lower bounds on what the AMM charges for thinner tokens, where its relative advantage in *providing* liquidity is greatest. Third, our MEV detection is deliberately conservative: the same-sender, opposite-side, wrapped-victim rule will miss sandwiches executed through distinct addresses or via private orderflow, so the ≈ 0.28 bps victim-impact estimate is a floor, not a census, of extraction. None of these limitations is fixable by simulation, and we have not tried; each is a direct consequence of insisting that every number trace to a live API or an on-chain log. We regard that constraint as the paper’s

central discipline.

11 Conclusion

Using only real data—live order books, decoded on-chain swaps, and aggregator quotes—we compare four market mechanisms trading the same asset at the same time. The evidence is clear and largely consistent with classical microstructure theory: the centralized order book prices information first and quotes the tightest spread; the on-chain order book tracks it almost perfectly at ten times the cost; the AMM offers a mechanically wider but still deep market whose price is dragged into line by sub-minute arbitrage; and the aggregator delivers the best on-chain execution, weakly dominating any single pool. AMM-specific frictions are real but bounded: flow concentrates in the cheapest fee tier, and detectable MEV in the deepest pool adds well under a basis point. The JELLY governance crisis, which we can study only because blockchain data are permanent, left no measurable footprint on the broader ETH market. Two implications follow. For *market design*, the AMM’s competitive advantage is not price—order books dominate on cost for liquid assets—but permissionless, always-on, composable liquidity; its equilibrium niche is the long tail of assets and the demand for on-chain settlement, not the deepest markets. For *research*, the immutability of on-chain data is a methodological asset that, paired with the ephemerality of exchange APIs, both constrains and enables credible natural experiments. We release all code and data for replication.

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Appendix A. Additional Details and Robustness

A.1 Data provenance. Table 1 lists every series with its exact observation count. The synchronized window (5,048 minutes) is bounded by Hyperliquid’s candle retention; Gate.io’s 1-minute history is longer (≈ 6.9 days) and is truncated to the common window for the aligned analyses. On-chain swaps were collected in 5,000-block batches via `eth_getLogs` with automatic fallback to 2,000-block sub-batches on RPC failure; no block ranges were lost.

A.2 Documented data gaps. Two API limits bind. (i) Hyperliquid candles retain ≈ 180 days, so ETH candles for the March 2025 JELLY window are unavailable; the JELLY DiD consequently uses Uniswap (permanent on-chain) as treatment and Gate.io daily candles as control. (ii) Gate.io rejects candle requests more than 10,000 points in the past, so hourly candles for the JELLY window are unavailable; we use daily candles, which are retained. We did not substitute any simulated value for these gaps.

A.3 Robustness of the AMM effective spread. Effective spreads are winsorized at the 99.5th percentile; the mean (median) is 11.05 (10.10) bps and is stable to reasonable trimming thresholds because the distribution is tight (std 2.26). The pool-mid proxy—the previous swap’s post-trade price—is the theoretically correct pre-trade mid for the next swap under continuous trading.

A.4 Half-life estimation. AR(1) coefficients are estimated by OLS on the minute-level deviation series; both are well inside the unit circle ($\phi_{\text{HL}} = 0.70$, $\phi_{\text{Uni}} = 0.42$), so the implied half-lives $-\ln 2 / \ln \phi$ are well defined.

A.5 Reproducibility. The archive contains six scripts (01–06) that collect, process, analyze, and plot; a machine-readable `analysis_results.json`; and all intermediate CSVs. Re-running the pipeline against the live APIs reproduces the qualitative results, subject to

the moving 180-day/10,000-point history windows.